

GARBITA CHOWDHURY

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Profile Summary

B.Tech CSE Student & Full-Stack Web Developer | Winner, SIH Internal Hackathon | WBJEE Rank 1102. Computer Science student at **Kalyani Govt. Engineering College (KGEC)** with expertise across Full-Stack Web development including **TypeScript, Next.js, NextAuth (JWT), React, Docker, Tailwind CSS, MongoDB, Vercel, HTML5, Node.js, and Python**. Skilled in Graphic Designing, Video Editing, Generative AI, Problem Solving, and Public Speaking.

Education

Kalyani Government Engineering College (KGEC) — B.Tech in CSE (Semester 3) 2026 – Present	YGPA: 8.16 WBJEE Rank: 1102
Bidya Bharati Girls' High School — Higher Secondary (Class XII - Science) 2023 – 2025	1st Division (Grade A)
Bidya Bharati School — Secondary (Class X) 2023	1st Division (Grade A)

Technical Skills & Competencies

Core & Creative: Full-Stack Web Development, DSA, Graphic Design, Video Editing, UI/UX
Languages & DB: TypeScript, JavaScript (ES6+), Python, HTML5, CSS3, C, SQL, MongoDB, PostgreSQL, SQLite
Frameworks & Web: Next.js, NextAuth.js (JWT), React.js, Tailwind CSS, Node.js, Express.js, Zustand, Leaflet.js, REST APIs
Tools & Hosting: Vercel, Docker, Git, GitHub, VS Code, Linux, Canva, Premiere Pro, Postman | Problem Solving, Decision Making, Public Speaking

Key Projects

- 1. NER-SHIELD: AI/ML Landslide Early Warning System (SIH 2026 Winner - ML, Web & Flutter App)**
- **ML Risk Engine:** Engineered a Random Forest classifier trained on 34 telemetry parameters (9,866 in-situ records), delivering 98.33% risk prediction accuracy and 6–24-hour advance hazard warnings.
 - **Web & Flutter App:** Developed an interactive Leaflet GIS Web App (React, Next.js, Tailwind) & cross-platform Flutter Mobile App (offline SQLite storage) paired with FastAPI + TextBee SMS emergency gateway for zero-internet alert dispatch.
 - **Technologies:** Python, Machine Learning, TypeScript, Next.js, React.js, Flutter, SQLite, Tailwind CSS, Leaflet.js, FastAPI, TextBee SMS.
- 2. ShopTrend: Next.js & NextAuth E-Commerce Storefront (Full-Stack) [github link](#) [app link](#)**
- **Features:** Full-stack e-commerce platform featuring secure JWT user authentication via NextAuth.js, MongoDB product catalog integration, Cloudinary image management, dynamic category filtering, persistent Zustand shopping cart state, and a full admin panel hosted on Vercel.
 - **Performance & UI:** Engineered fast server rendered React components ensuring instant product catalog rendering, responsive UI, and protected user checkout sessions.
 - **Technologies:** TypeScript, Next.js, NextAuth.js (JWT), React.js, Tailwind CSS, Zustand, MongoDB, Node.js, REST APIs
- 2. WeatherTrack: Next.js & Integrated Task Planner Web App (Full-Stack) [github link](#) [app link](#)**
- **Web App & Features:** Full-stack Next.js single-page weather dashboard featuring an interactive OpenStreetMap background integration, real-time Open-Meteo meteorological telemetry, floating glassmorphic forecast overlays, and a calendar task planner with MongoDB state persistence (To Do, Ongoing, Completed).
 - **Performance & UI:** Designed a borderless glassmorphic UI with automatic GPS location connection, live UV index & atmospheric pressure toaster alerts, and fast server-rendered UI hosted on Vercel.
 - **Technologies:** TypeScript, Next.js, React.js, Tailwind CSS, MongoDB, Open-Meteo REST API, OpenStreetMap, Vercel.

Achievements & Honors

- **Winner, SIH Internal Campus Hackathon (SIH 2026):** Secured 1st Place with Team SiXfinity (Problem Statement 26001), advancing to national evaluation rounds.
- **WBJEE Rank 1102:** Achieved top merit rank among 100,000+ engineering candidates in West Bengal.
- **Academic Excellence:** Awarded Grade 'A' (1st Division) in Secondary (Class X) & Higher Secondary (Class XII).
- **Vigyan Abhijan 2026:** Active event coordinator, public speaker, and video editing contributor in campus technology competitions.